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A rectangular matrix is in **row echelon form** if it has the following three properties:
(1) All nonzero rows are above any rows of all zeroes.
(2) Each leding entry of a row is in a column to the right of the leading entry of the row above it.
(3) All entries in a column below a leading entry are zeroes.
If a matrix in echelon form satisfies the following additional conditions then it is in **reduced row echelon form**:
(4) The leading entry in each nonzero row is 1
(5) Each leading 1 is the only nonzero entry in its column

A **pivot position** in a matrix A is a location in A that corresponds to a leading 1 in the reduced echelon form of A. A **pivot column** is a column of A that contains a pivot position.

Existence and Uniqueness Theorem:
A linear system is consistent if and only if the rightmost column of the augmented matrix is *not* a pivot column - that is, if and only if an echelon form of the augmented matrix has *no* rows of the form

$$[0 \ ... \ 0 \ \ b] \text{ with } b \text{ nonzero}$$

If a linear system is consistent, then the solution set contains either (i) a unique solution, when there are no free variables or (ii) infinitely many solutions, when there is at least one free variable.

If $\mathbf{v_1}, ..., \mathbf{v_p}$ are in $\mathbb{R}^n$, then the set of all linear combinations of $\mathbf{v_1}, ..., \mathbf{v_p}$ is denoted by Span$\{v_1, ..., v_p\}$ and is called the **subset of $\mathbb{R}^n$ spanned (or generated) by** $v_1, ..., v_p$. That is, Span$\{v_1, ..., v_p\}$ is the collection of all vectors that can be written in the form

$$c_1v_1 + c_2v_2 + ... + c_pv_p$$

with $c_1, ..., c_p$ scalars.

Let A be an $m \times n$ matrix. Then the following statements are logically equivalent. That is, for a particular A, either they are all true statements or they are all false.
a. For each b in $\mathbb{R}^m$, the equation $Ax = b$ has a solution.
b. Each b in $\mathbb{R}^m$ is a linear combination of the columns of A.
c. The columns of A span $\mathbb{R}^m$.
d. A has a pivot position in every row.

An indexded set of vectors $\mathbf{v_1}, ..., \mathbf{v_p}$ in $\mathbb{R}^n$ is said to be **linearly independent** if the vector equation

$$x_1v_1 + x_2v_2 + ... + x_pv_p = 0$$

has only the trivial solution. The set $\mathbf{v_1}, ..., \mathbf{v_p}$ is said to be **linearly dependent** if there exist weights $c_1, c_2, ..., c_p$ not all 0 such that

$$c_1v_1 + c_2v_2 + ... + c_pv_p = 0$$

Let $T : \mathbb{R}^n \to R^m$ be a linear transformation and let A be the standard matrix for T. Then:
a. T maps $\mathbb{R}^n$ onto $\mathbb{R}^m$ if and only if the columns of A span $\mathbb{R}^m$.
b. T is one-to-one if and only if the columns of A are linearly independent.

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. If $ad - bc \neq 0$, then A is invertible and

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

If $ad - bc = 0$, then A is not invertible.

Let A and B denote matrices whose sizes are appropriate for the following sums and products.
a. $(A^T)^T = A$
b. $(A + B)^T = A^T + B^T$
c. For any scalar r, $(rA)^T = rA^T$
d. $(AB)^T = B^TA^T$

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a. If A is an invertible matrix, then A^{-1} is invertible and

$$(A^{-1})^{-1} = A$$

b. If A and B are $n \times n$ invertible matrices, then so is AB, and the inverse of AB is the product of inverses of A and B in the reverse order. That is,

$$(AB)^{-1} = B^{-1}A^{-1}$$

c. If A is an invertible matrix, then so is A^T and the inverse of A^T is the transpose of A^{-1}. That is,

$$(A^T)^{-1} = (A^{-1})^T$$

The determinant of an $n \times n$ matrix A can be computed by a cofactor expansion across any row or down any column. The expansion across the ith row using the cofactors is

$$\det A = a_{i1}C_{i1} + a_{i2}C_{i2} + ... + a_{in}C_{in}$$

The cofactor expansion down the jth column is

$$\det A = a_{1j}C_{1j} + a_{2j}C_{2j} + ... + a_{nj}C_{nj}$$

Row Operations for Determinants
Let A be a square matrix.
a. If a multiple of one row of A is added to another row to produce a matrix B, then det B = det A.
b. If two rows of A are interchanged to produce B, then det $B = -$det A.
c. If one row of A is multiplied by k to produce B, then det $B = k \cdot$ det A.

Cramer's Rule
Let A be an invertible $n \times n$ matrix. For any b in $\mathbb{R}^n$, the unique solution of $Ax = b$ has entries given by

$$x_i = \frac{\det A_i(b)}{\det A}, \; i = 1, 2, ..., n$$

Adjugate Matrix/Inverse
Say A is an invertible $n \times n$ matrix. We have

$$\text{adj } A = \left[C_{ji}\right]$$

where C_{ij} is the ij-cofactor and then inverse formula

$$A^{-1} = \frac{1}{\det A} \text{adj } A$$

A **subspace** of a vector space V is a subset H of V that has three properties:
a. The 0 vector of V is in H.
b. H is closed under vector addition. That is, for each u and v in H, the sum $u + v$ is in H.
c. H is closed under multiplication by scalars. That is, for each u in H and each scalar c, the vector cu is in H.

The **null space** of an $m \times n$ matrix A, written Nul A is the set of all solutions of the homogeneous equation $Ax = 0$. In set notation,

$$\text{Nul } A = \{x : x \text{ is in } \mathbb{R}^n \text{ and } Ax = 0\}$$

The **column space** of an $m \times n$ matrix A, written as Col A, is the set of all linear combinations of the columns of A. If $A = [a_1 ... a_n]$, then

$$\text{Col } A = \text{Span}\{a_1, ..., a_n\}$$

Let H be a subspace of a vector space V. A set of vectors B in V is a **basis** for H if
(i) B is a linearly independent set, and
(ii) The subspace spanned by B coincides with H; that is,

$$H = \text{Span } B$$

A **linear transformation** T from a vector space V into a vector space W is a rule that assigns to each vector x in V a unique vector $T(x)$ in W such that
(i) $T(u + v) = T(u) + T(v)$ for all $u, v \in V$, and
(ii) $T(cu) = cT(u)$ for all u in V and scalars c.

The Invertible Matrix Theorem
Let A be a square $n \times n$ matrix. Then the following statements are equivalent. That is, for a given A, the statements are all true or all false.
a. A is an invertible matrix.
b. A is row equivalent to the $n \times n$ identity matrix.
c. A has n pivot positions.
d. The equation $Ax = 0$ has only the trivial solution.
e. The columns of A form a linearly independent set.
f. The linear transformation $x \to Ax$ is one-to-one.
g. The equation $Ax = b$ has at least one solution for each b in $\mathbb{R}^n$
h. The columns of A span $\mathbb{R}^n$
i. The linear transformation $x \to Ax$ maps $\mathbb{R}^n$ onto $\mathbb{R}^n$.
j. There is an $n \times n$ matrix C such that $CA = I$.
k. There is an $n \times n$ matrix D such that $AD = I$.
l. A^T is an invertible matrix.

A **vector space** is a nonempty set V of objects called *vectors* on which there are two defined operations called *addition* and *multiplication by scalars* (real numbers), subject to the ten axioms listed below. The axioms must hold for all vectors u, v, and w in V and for all scalars c and d.
1. The sum of u and v, denoted $u + v$, is in V.
2. $u + v = v + u$
3. $(u + v) + w = u + (v + w)$
4. There is a **zero** vector 0 in V such that $u + 0 = u$
5. For each u in V, there is a vector $-u$ in V such that $u + (-u) = 0$.
6. The scalar multiple of u by c, denoted cu, is in V.
7. $c(u + v) = cu + cv$
8. $(c + d)u = cu + du$
9. $c(du) = (cd)u$
10. $1u = u$

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The Unique Representation Theorem:

Let $\mathcal{B} = \{b_1, ..., b_n\}$ be a basis for a vector space V. Then for each $x \in V$ there exists a unique set of scalars $c_1, ..., c_n$ such that $x = c_1 b_1 + ... + c_n b_n$

If a vector space V has a basis of n vectors, then every basis of V must consist of exactly n vectors.

The Basis Theorem:

Let V be a p-dimensional vector space, $p \geq 1$. Any linearly independent set of exactly P elements in V is automatically a basis for V. Any set of exactly p elements that spans V is automatically a basis for V.

The Rank Theorem:

The dimensions of the column space and the null space of an $m \times n$ matrix A satisfy the equation rank $A +$ nullity $A = n$

For $\mathcal{B} = \{b_1, ..., b_n\}$, $\mathcal{C} = \{c_1, ..., c_n\}$ bases for vector space V there is unique $n \times n$ matrix $\underset{\mathcal{C} \leftarrow \mathcal{B}}{P}$ for which $[x]_{\mathcal{C}} = \underset{\mathcal{C} \leftarrow \mathcal{B}}{P}[x]_{\mathcal{B}}$, that P being $\underset{\mathcal{C} \leftarrow \mathcal{B}}{P} = \begin{bmatrix} [b_1]_{\mathcal{C}} & [b_2]_{\mathcal{C}} & ... & [b_n]_{\mathcal{C}} \end{bmatrix}$

An **eigenvector** of an $n \times n$ matrix A is a nonzero vector x such that $Ax = \lambda x$ for some scalar λ. A scalar λ is an **eigenvalue** of A if there is a nontrivial solution x of $Ax = \lambda x$, such an x is an eignvector corresponding to λ.

If $v_1, ..., v_r$ are eigenvectors corresponding to distinct eigenvalues, then the set $\{v_1, ..., v_r\}$ is linearly independent.

A scalar λ is an eigenvalue of an $n \times n$ matrix A if and only if λ satisfies the **characteristic equation** $\det(A - \lambda I) = 0$

Two $n \times n$ matrices A and B are **similar** if there exists an invertible matrix P for which $A = PBP^{-1}$

Two similar matrices have the same characteristic polynomial and the same eigenvalues.

The Diagonalization Theorem:

An $n \times n$ real matrix A is diagonalizable over $\mathbb{R}$ if and only if A has n linearly independent eigenvectors, via $A = PDP^{-1}$ with the columns of P being the n linearly independent eigenvectors of A and D the eigenvalues of A, in the same order.

Say A is an $n \times n$ matrix with distinct eigenvalues $\lambda_1, \lambda_2, ..., \lambda_p$. Then the dimension of the eigenspace is less than or equal to its multiplicity in the characteristic polynomial. And A is diagonalizable over $\mathbb{R}$ if and only if the sum of the dimeinsions of the eigenspaces equals n, i.e. if the dimension of the eigenspace of each λ_k equals its multiplicity in the characteristic polynomial.

For V a vector space, an eigenvector of a linear transformation $T : V \to V$ is a nonzero vector x in V for which $T(x) = \lambda x$ for some scalar λ, λ being an eigenvalue of T.

The matrix M of a linear transformation T satisfies $[T(x)]_{\mathcal{B}} = M[x]_{\mathcal{B}}$ with $M = \begin{bmatrix} [T(b_1)]_{\mathcal{B}} & [T(b_2)]_{\mathcal{B}} & ... & [T(b_n)]_{\mathcal{B}} \end{bmatrix}$

For A a 2×2 real matrix with a complex eigenvalue $\lambda = a - bi$, $b \neq 0$, and associated eigenvector $v \in \mathbb{C}^2$ then $A = PCP^{-1}$ with $P = \begin{bmatrix} \text{Re } v & \text{Im } v \end{bmatrix}$ and $C = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$

For $u = (u_1, ..., u_n)$, $v = (v_1, ..., v_n)$ vectors in $\mathbb{R}^n$, the **dot product** of u and v, is $u \cdot v = u_1 v_1 + ... + u_n v_n$

In $\mathbb{R}^n$, the length (norm) of $v = (v_1, ..., v_n)$, $||v||$, is given by $||v|| = \sqrt{v_1^2 + ... + v_n^2}$ with $||v||^2 = v \cdot v$

The **distance** between two vectors u and v in $\mathbb{R}^n$ is given by $||u - v||$

Two vectors u and v in $\mathbb{R}^n$ are **orthogonal** if $u \cdot v = 0$

The **orthogonal complement** of a subspace $W^{\perp}$ of $\mathbb{R}^n$ is the set of all vectors in $\mathbb{R}^n$ which are orthogonal to W, i.e. orthogonal to each vector in W. $W^{\perp}$ is always a subspace of $\mathbb{R}^n$

For A any matrix we have $(\text{Row } A)^{\perp} = \text{Nul } A$ and $(\text{Col } A)^{\perp} = \text{Nul } A^T$

An **orthogonal set** of vectors is a set of vectors in which each pair of distinct vectors are orthogonal to each other. A set of nonzero orthogonal vectors is always linearly independent. An **orthogonal basis** for a subspace W of $\mathbb{R}^n$ is a basis which is also an orthogonal set.

For an orthogonal basis $\{u_1, ..., u_p\}$ of a subspace W of $\mathbb{R}^n$ then for writing an aribtrary $y \in W$ in the form $y = c_1 u_1 + ... + c_p u_p$, we have $c_j = \dfrac{y \cdot u_j}{u_j \cdot u_j}$ for each $1 \leq j \leq p$

For each nonzero u in $\mathbb{R}^n$ we can write any y in the form $y = \hat{y} + z$ where $\hat{y}$ is a multiple of u and z is orthogonal to u, $\hat{y}$ is the **orthogonal projection of y onto u** and z is the **component of y orthogonal to u**.

An **orthonormal set** is an orthogonal set of unit vectors (length one)

An $m \times n$ matrix U has orthonormal columns if and only if $U^T U = I_n$. Such a matrix always satisfies $||Ux|| = ||x||$

A square matrix U is an **orthogonal matrix** if it has orthonormal columns, i.e. if $U^{-1} = U^T$

The Orthogonal Decomposition Theorem:

For W a subspace of $\mathbb{R}^n$, each $y \in R^n$ can be written uniquely in the form $y = \hat{y} + z$ where $\hat{y} \in W$ and $z \in W^\perp$ where $\hat{y}$ is the **orthogonal projection of y onto W**, also denoted $\text{proj}_W y$. And if $\{u_1, ..., u_p\}$ is an orthogonal basis for W then $\hat{y} = \dfrac{y \cdot u_1}{u_1 \cdot u_1} u_1 + ... + \dfrac{y \cdot u_p}{u_p \cdot u_p} u_p$

The Best Approximation Theorem:

For W a subspace of $\mathbb{R}^n$, for any $y \in R^n$, $\hat{y}$ is the closest point in W to y, i.e. $||y - \hat{y}|| < ||y - w||$ for any $w \in W$ for which $w \neq \hat{y}$

For $\{u_1, ..., u_p\}$ an orthonormal basis for W a subspace of $\mathbb{R}^n$, then $\text{proj}_W y = (y \cdot u_1)u_1 + ... + (y \cdot u_p)u_p$ and if U is the matrix whose columns are the u_i''s then $\text{proj}_W y = UU^T y$

The Gram-Schmidt Process:

For $\{x_1, ..., x_p\}$ a basis for a nonzero subspace W of $\mathbb{R}^n$ iteratively defining for each $1 \leq k \leq p$, $v_k = x_k - \sum_{i=1}^{k-1} \dfrac{x_k \cdot v_i}{v_i \cdot v_i} v_i$ then $\{v_1, ..., v_p\}$ will be an orthogonal basis for W and for each such k $\text{Span}\{v_1, ..., v_k\} = \text{Span}\{x_1, ..., x_k\}$

The QR Factorization:

If A is an $m \times n$ matrix with linearly independent columns, then can write $A = QR$ where Q is $m \times n$ matrix whose columns are an orthonormal basis for Col A and R is an $n \times n$ upper triangular matrix, all of whose diagonal entries are positive.

For A an $m \times n$ matrix, $b \in R^m$ a **least-squares solution** of $Ax = b$ is $\hat{x} \in \mathbb{R}^n$ such that $||b - A\hat{x}|| \leq ||b - Ax||$ for all $x \in \mathbb{R}^n$.

The set of least squares solution of $Ax = b$ is the nonempty set of solutions to the equation $A^T Ax = A^T b$

For A an $m \times n$ matrix, these statements are equivalent:
(1) The equation $Ax = b$ has a unique least squares solution for each b
(2) The columns of A are linearly independent
(3) $A^T A$ is invertible.
When any of these are true, the least squares solution is given by $\hat{x} = (A^T A)^{-1} A^T b$

When A has linearly independent columns, can also use QR factorization to write the unique least-squares solution as $\hat{x} = R^{-1} Q^T b$

An **inner product** on a vector space V assigns to each pair of vectors a real number, denoted $\langle u, v \rangle$, satisfying the axioms for all vectors u, v, w and all scalars c.
(1) $\langle u, v \rangle = \langle v, u \rangle$
(2) $\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$
(3) $\langle cu, v \rangle = c\langle u, v \rangle$
(4) $\langle u, u \rangle \geq 0$ with $\langle u, u \rangle = 0$ only for $u = 0$

In any inner product space $||v|| = \sqrt{\langle v, v \rangle}$, unit vector has length 1, and u, v orthogonal if $\langle u, v \rangle \ = 0$

Cauchy-Schwarz Inequality:

In any inner product space $|\langle u, v \rangle| \leq ||u|| \ ||v||$

Triangle Inequality:

In any inner product space, $||u + v|| \leq ||u|| + ||v||$

A square matrix is **symmetric** if $A = A^T$. In a real symmetric matrix any two eigenvectors from different eigenvalues are orthogonal.

A matrix is **orthogonally diagonalizable** if it can be written in the form PDP^{-1} where P is orthogonal and D diagonal. A matrix is orthogonally diagonalizable if and only if it is symmetric.

The Spectral Theorem:

If A is a real symmetric $n \times n$ then A has n eigenvalues counting multiplicities and the dimension of each eigenspace is equal to the multiplicity of that eigenvalue in the characteristic equation. The eigenspaces are mutually orthogonal and A is orthogonally diagonalizable.

Spectral decomposition is writing $A = \lambda_1 u_1 u_1^T + ... + \lambda_n u_n u_n^T$ where the u_i's are orthonormal eigenvectors and the λ_i's are the corresponding eigenvalues.

For A a symmetric $n \times n$ matrix and $x = \begin{bmatrix} x_1 & ... & x_n \end{bmatrix}$, $Q = x^T Ax$ is a **quadratic form**.

For A $n \times n$ symmetric, there is an orthogonal change of variable $x = Py$ that transforms the quadratic form $x^T Ax$ into a quadratic form $y^T Dy$ with no cross-product term, D a diagonal matrix.

A quadratic form is **positive definite** if $Q(x) > 0$ for all $x \neq 0$, negative definite if $Q < 0$, indefinite if mixed. Quadratic form is positive definite if and only if all eigenvalues of associated A are positive. Form is **positive semidefinite** if $Q(x) \geq 0$ for all x which only happens only when all eigenvalues of A are nonnegative.

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The **singular value decomposition** of any $m \times n$ matrix A is expressing it in the form $U \sum V^T$ where U is an orthogonal $m \times m$ matrix, $\sum$ is an $m \times n$ "diagonal" matrix, and V is an orthogonal $n \times n$ matrix.

$S = A^T A$ is a symmetric matrix with nonnegative real eigenvalues $\lambda_1, \lambda_2, ..., \lambda_n$ whose eigenvectors $v_1, v_2, ..., v_n$ form an orthonormal basis of $\mathbb{R}^n$, and collectively form the matrix V. The **singular values** of A are $\sigma_1, \sigma_2, ..., \sigma_n$ for which $\sigma_j = \sqrt{\lambda_j}$, these being in decreasing order. Taking $u_i = \dfrac{1}{\sigma_i} A v_i$ for each i such that $\sigma_i \neq 0$ and then filling out the u_i's to an orthonormal basis for $\mathbb{R}^m$ using Gram-Schmidt as needed, we form the matrix U.

Then $\sum$ is the "diagonal" matrix whose diagonal entries are the singular values of A in decreasing order.

Differential Equations sections start here

Existence and Uniqueness: Homogeneous Case

For any real number $a \neq 0$, b, c, t_0, Y_0, Y_1, there exista a unique solution to the initial value problem

$ay'' + by' + cy = 0$; $y(t_0) = Y_0$, $y'(t_0) = Y_1$

Linear Independence of Two Functions

A pair of functions $y_1(t)$ and $y_2(t)$ is said to be **linearly independent** on the interval I if and only if neither of them is a constant multiple of the other on all of I. We say they are **linearly dependent on** I if one of them is a constant multiple of the other on all of I

Representation of Solutions to Initial Value Problem

If $y_1(t)$ and $y_2(t)$ are any two solutions to the homogeneous differential equation that are linearly independent on $\mathbb{R}$, then unique constants c_1 and c_2 can always be found so that $c_1 y_1(t) + c_2 y_2(t)$ satisfies the initial value problem listed in the homogeneous case.

A Condition for Linear Dependence of Solutions

For any real numbers $a \neq 0$, b, and c, if $y_1(t)$ and $y_2(t)$ are any two solutions to the homogeneous differential equation on $\mathbb{R}$ and if the equality
$y_1(\tau) y_2'(\tau) - y_1'(\tau) y_2(\tau) = 0$
holds at any point τ then y_1 and y_2 are linearly dependent on $\mathbb{R}$. The expression on the left hand side of this is called the *Wronskian* of y_1 and y_2.

For $ay'' + by' + cy = 0$, the equation $ar^2 + br + c = 0$ is the characteristic (auxiliary) equation. If it has distinct real roots r_1, r_2 then both $y_1(t) = e^{r_1 t}$, $y_2(t) = e^{r_2 t}$ are solutions and $y(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}$ is the general solution.
If this characteristic equation has a repeated root r, then the general solution is $y(t) = c_1 e^{rt} + c_2 t e^{rt}$

Complex Conjugate Roots

If the characteristic equation of the above has complex conjugate roots $\alpha \pm i\beta$, then the general solution is $y(t) = c_1 e^{\alpha t} \cos(\beta t) + c_2 e^{\alpha t} \sin(\beta t)$

Method of Undetermined Coefficients

To find a particular solution of $ay'' + by' + cy = Ct^m e^{rt}$ where m is a nonnegative integer use the form
$y_p(t) = t^s (A_m t^m + ... + A_1 t + A_0) e^{rt}$ with
(i) $s = 0$ if r is not a root of the associated characteristic equation.
(ii) $s = 1$ if r is a single root of the associated characteristic equation.
(iii) $s = 2$ if r is a double root of the associated characteristic equation.

To find a particular solution of $ay'' + by' + cy = Ct^m e^{\alpha t} \cos(\beta t)$ or $Ct^m e^{\alpha t} \sin(\beta t)$ for $\beta \neq 0$ where m is a nonnegative integer use the form
$y_p(t) = t^s (A_m t^m + ... + A_1 t + A_0) e^{\alpha t} \cos(\beta t) +$
$t^s (B_m t^m + ... + B_1 t + B_0) e^{\alpha t} \sin(\beta t)$ with
(i) $s = 0$ if $\alpha + i\beta$ is not a root of the associated characteristic equation.
(ii) $s = 1$ if $\alpha + i\beta$ is a root of the associated characteristic equation.

Superposition Principle

If y_1 is a solution to $ay'' + by' + cy = f_1(t)$ and y_2 to $ay'' + by' + cy = f_2(t)$ then for any constants k_1 and k_2 the function $k_1 y_1 + k_2 y_2$ is a solution to
$ay'' + by' + cy = k_1 f_1(t) + k_2 f_2(t)$

Existence and Uniqueness: Nonhomogeneous Case

For any real numbers $a \neq 0$, b, c, t_0, Y_0, Y_1 then there is a unique solution to
$ay'' + by' + cy = f(t)$, $y(t_0) = Y_0$, $y'(t_0) = Y_1$
of the form $y(t) = y_p + c_1 y_1 + c_2 y_2$ where y_p is a particular solution and y_1 and y_2 are solutions to the homogeneous equation.

Variation of Parameters

To determine a particular solution of $ay'' + by' + cy = f$
(a) Find two linearly independent solution y_1, y_2 to homogeneous equation and take $y_p = u_1 y_1 + u_2 y_2$
(b) Determine u_1 and u_2 via

$$u_1' = \frac{\begin{vmatrix} 0 & y_2 \\ \frac{f(t)}{a} & y_2' \end{vmatrix}}{\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}}, \quad u_2' = \frac{\begin{vmatrix} y_1 & 0 \\ y_1' & \frac{f(t)}{a} \end{vmatrix}}{\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}}$$

System of differential equations expressed by
$x_1' = a_{11}(t)x_1 + a_{12}(t)x_2 + ... + a_{1n}(t)x_n$
$x_2' = a_{21}(t)x_1 + a_{22}(t)x_2 + ... + a_{2n}(t)x_n$ ...
$x_n' = a_{n1}(t)x_1 + a_{n2}(t)x_2 + ... + a_{nn}(t)x_n$
is a linear homogeneous system in **normal form**.

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n Linearly Independent Eigenvectors

If A has n linearly independent eigenvectors $u_1, ..., u_n$ and r_i the eigenvalue for u_i then
$\{e^{r_1 t}u_1, e^{r_2 t}u_2, ..., e^{r_n t}u_n\}$
is a fundamental solution set and
$X(t) = [e^{r_1 t}u_1 \ e^{r_2 t}u_2 \ ... \ e^{r_n t}u_n]$ is a fundamental matrix so that a general solution to $x' = Ax$ is
$x(t) = c_1 e^{r_1 t}u_1 + c_2 e^{r_2 t}u_2 + ... + c_n e^{r_n t}u_n$

Complex Eigenvalues

If real matrix A has eignevalues $\alpha \pm i\beta$ with corresponding eigenvectors $a \pm ib$ then two linearly independent real vector solutions to $x'(t) = Ax(t)$ are
$e^{\alpha t}\cos(\beta t)a - e^{\alpha t}\sin(\beta t)b$
$e^{\alpha t}\sin(\beta t)a + e^{\alpha t}\cos(\beta t)b$

Fourier Series

If f is piecewise continuous on $[-L, L]$ then we have

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left\{a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right)\right\}$$

where $a_n = \dfrac{1}{L}\displaystyle\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)\,dx$

and $b_n = \dfrac{1}{L}\displaystyle\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx$

Orthogonal System

is a set of functions $\{f_n(x)\}$ with respect to the nonnegative weight function $w(x)$ on $[a, b]$ if

$$\int_a^b f_m(x)f_n(x)w(x)\,dx = 0 \text{ for all } m \neq n$$

For f continuous and periodic of period $2L$ and f' piecewise continuous on $[-L, L]$ then the Fourier series for f converges *uniformly* and we can obtain the Fourier series for f' and $\displaystyle\int f$ via term by term differentiation/integration.

Fourier Cosine and Sine Series

For f piecewise continuous on $[0, L]$ the **Fourier cosine series** on this interval is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right)$$

where $a_n = \dfrac{2}{L}\displaystyle\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right)\,dx$

and the sin series is

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

where $b_n = \dfrac{2}{L}\displaystyle\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx$

Trig Formulas

$\sin(A + B) = \sin A\cos B + \cos A\sin B$
$\sin(A - B) = \sin A\cos B - \cos A\sin B$
$\cos(A + B) = \cos A\cos B - \sin A\sin B$
$\cos(A - B) = \cos A\cos B + \sin A\sin B$
$\sin(2A) = 2\sin A\cos A$
$\cos(2A) = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$
$\sin^2 A + \cos^2 A = 1$
$\sec^2 A - \tan^2 A = 1$

Trig Integrals

$\displaystyle\int \sin x\,dx = -\cos x$

$\displaystyle\int \tan x\,dx = \ln|\sec x|$

$\displaystyle\int \sec^2 x\,dx = \tan x$

$\displaystyle\int \sec x\tan x\,dx = \sec x$

$\displaystyle\int \sec x\,dx = \ln|\sec x + \tan x|$